

CHAPTER 2.B - YOUR HOME IS A HOLON WAITING TO BE KISSED AWAKE

Once upon a time, there was a kingdom that fell asleep because it was put under a spell. There was no jealous queen, no enchanted apple, no spindle that pricked a careless finger. The spell that put the electricity system to sleep for well over a century, was cast by a few hundred earnest engineers working around turn of the 19th century. They were ordinary men and women working in ordinary offices in Berlin and Pittsburgh and Manchester. If you had visited them, the only thing that would have caught your eye is that their offices were remarkably well lit for the time. They didn't consider themselves wizards. They were just implementing electricity grids based on what was an entirely reasonable design choice at the time. The electricity system of the day had only one knob to turn -- how much fuel was burned centrally -- so they built a system that turned that one knob extremely well.

That system with that one knob worked beautifully for over a century. It worked for their children, their grandchildren and even the grandchildren of their grandchildren. And as the grid kept working -- essentially unchanged -- everyone forgot that other knobs existed that could be turned. These other knobs have become extremely relevant now that we are replacing fossil fuels with solar, wind, batteries and grid-forming inverters. Some people in the kingdom are waking up to that new reality. Unfortunately the king (the fossil energy producer) and queen (the grid operator) are still deeply asleep.

In regular fairy tales a handsome prince arrives, kisses a beautiful princess, and the spell breaks. But, quite apart from the gender issues, this is, as you may have already suspected, not how it is going to work for the electricity system. The king and queen seem intent on sleeping a few decades longer. Instead the kingdom will have to be kissed awake one building at a time. Kissing a building awake is done by attaching batteries, grid-forming inverters and optionally solar panels and electric vehicles to the building. Thus activated the building becomes a little semi-autonomous kingdom of its own. A kingdom that can now provide its inhabitants with resilient, affordable and clean energy. The little kingdom becomes a safe energy refuge for its inhabitants, even when the kingdoms around it fail.

This is the chapter where we stop talking and start kissing.

Home Holons are Waking Up All Over the World

Amara's home

Remember Amara? She lives in rural Africa and cooks on wood and dried dung that she gathers during the day the way her mother taught her. It's labour intensive, the smoke is poisoning her lungs, it's harder and harder to find wood and dung, but it's the only fuel she can afford. In the evening her children do their homework while breathing in the poisonous fumes of tiny kerosene lights.

Amara is not alone. This is a reality for a billion people worldwide. Billions more are living in extreme energy poverty, at least compared to the majority of the people reading this book.

Amara's prince happens to be a woman and she doesn't arrive to a gala on a white horse but on a cheap electric bicycle on a hot and dusty Monday morning. Behind her bike is a small dinged up trailer containing her tools and a few solar panels, a battery, a cheap induction stove, a few LED lights, a phone charger and a satellite receptor. The woman screws the solar panels and satellite receptor onto the corrugated iron sheets on Amara's roof. Then she shows Amara how all this works and how Amara can pay back the small loan for the system with micropayments by phone.

No grid was extended. No central plan was approved. No minister cut a ribbon. No bank did an audit of Amara's credit rating. But this is possibly the largest moral upgrade per dollar available to our species.

The system is harsh: when Amara stops paying, the electricity stops. But the system is also simple and with very little overhead that could be corrupted and that Amara wouldn't be able to afford. Technological progress has given us the ability to offer Amara energy in a cheap and simple to use product that can be delivered using competitive markets. Amara now has access to dependable energy and information. Once she has paid off her loan, that energy will be free. The unassuming princess on her bike has awakened Amara's home and now it is providing her with not just shade and protection but also energy and information. It helps to provide Amara income, dignity, autonomy and human flourishing.

The customers nobody saw

At the turn of the millennium, a Chinese woman called Susan Li was visiting the slums and rural areas in India. She had just had children of her own, and when she saw all those Indian children suffering in extreme energy poverty she thought: not on my shift. In 2004 she founded Shenzhen Power Solutions, making products specifically aimed at what she calls *bottom of pyramid customers*. The phrase was coined a few years earlier by the Indian-born scholar C.K. Prahalad, who had noticed that about four billion people worldwide hardly appear in corporate spreadsheets or government policies because they are simply too poor and therefore too unimportant to matter. Aggregate the four billion, however, and you get the largest unserved market on Earth -- plus customers for whom every kilowatt-hour does life-altering work. Sources: [\(Prahalad & Hart 2002\)](#); [\(Wikipedia summary\)](#). [Prahalad primary]

I discovered all this because one of the products Susan Li sells caught my attention. It's the most basic version of the magic that Amara received in the previous section. It consists of four tiny solar cells, connected to a what looks like a yellow lunch box. The lunch box contains a small battery and inverter. Plugged into the lunch box with thin long wires are five LED lights. The LED lights are tiny compared to what I'm used to, but still much brighter than the kerosene lamp Amara uses. And the lunch box can also charge a mobile phone.

To me Susan Li is a bona fide heroine and it pains me that I live in a world where homes remain dark and children die from kerosene fumes because families don't have access to such a thirty dollar package. Source: ([CEIBS profile of Susan Li](#)).

Home holons are waking up everywhere

Fortunately it's not just Susan Li. Companies providing off grid electricity are popping up everywhere. 13 M-KOPA, founded in Nairobi in 2011, has installed more than five million pay-as-you-go solar systems across Kenya, Uganda, Nigeria, Ghana and South Africa and sixty-two per cent of its customers report the system generates income for them because it buys them light to work by, a charged phone to take orders on, a fridge to keep food from spoiling, and so on. It's a bridge from subsistence to enterprise achieved because of entrepreneurs willing to see those bottom of pyramid customers and because renewable energy technology is now able to deliver. Source: ([M-KOPA Impact](#)).

In Pakistan, the grid is both expensive and unreliable. Pakistani households have responded in the way most households would when they see a way out: they imported a whopping seventeen gigawatts of solar panels in 2024, and outdid that feat the next year by importing a truly mind boggling fifty gigawatts in 2025. Not because they didn't have a grid connection, but because they were sick and tired of the cost and unreliability of the one they had. [Pakistan 17 GW 2024 / 50 GW 2025 import figures against most recent customs / IEA / BloombergNEF data]

In South Africa the situation is similar. The grid operator there has been turning off entire neighbourhoods for hours at a stretch because there is not enough power to go around, so by now every homeowner who can afford it has put solar panels on their roof and batteries inside their homes and is no longer impacted by the outages. Source: ([PV Magazine: South Africa 2024](#)).

Compared to most grids, the Dutch grid is almost embarrassingly reliable. On average you will experience less than 1 day worth of outages in fifty years. Yet batteries also make sense for homes like mine. First of all because batteries allow me to make optimal use of my solar panels. That not just feels nice but it's also financially interesting because energy I sell back to the grid is worth just a fraction of what I have to pay when I buy electricity. (This mainly due to taxes.) With batteries I can run off my solar panels 90% of the time. On top of this my wife and I like the idea that in our home the light stay on when the unthinkable happens.

So you may not have noticed yet, but there is a revolution going on, with homes becoming reliable energy providers -- holons -- all over the world. And they don't require the blessing of big corporations or nation states. Just cheap renewable energy technology and people who see the opportunity and maybe give a damn.

Big Buildings Have Big Incentives

Most of what's true for small buildings like a home, is also true for bigger buildings. And the stakes increase if such a home cannot get enough grid capacity. This problem is especially

pronounced in the Netherlands where more and more companies have to wait many years for a (bigger) grid connection.

For such buildings in need of more energy from the grid, step one is to buy batteries so they can max out the grid connection they were able to get. Peak shaving can often double the energy they are able to get within their existing contract. This also gives them cheaper electricity because they can buy when the energy is cheap and optimise the use of their solar panels. Finally they now have an uninterrupted power supply so the power stays on for a while when the grid goes down. As batteries and grid forming inverters continue to become cheaper -- see the chapters on learning curves -- buildings will increasingly be equipped this way.

If the grid still can't provide energy, a diesel- or gas generator is needed. Where that is legal the business case for becoming a holon becomes even better because diesel generators produce very expensive electricity and you want to use as little of that as possible.

That's lot of larger buildings, also ready to wake up.

If I take a step back I feel that the ascend of building holons is already locked in. It's a survival kit for customers who cannot get a grid connection at all. It's a way to lower electricity costs for most other customers. It's increasingly a no-brainer for companies that cannot get enough electricity from the grid. And the costs keep plummeting. So the question is not whether building holons will ascend but how to use their abilities wisely, instead of denying they exist because of a spell that is now over a century old.

Your Friendly Neighbourhood Holon

But why stop at one building? Why not turn an entire neighbourhood into a holon? The local low-voltage transformer in a Dutch street serves around 50 to 250 households. Interestingly this is also a number sociologists will give you when you ask them how many neighbours you can feel close to without making an effort. A neighbourhood holon, as it happens, is a unit at which engineering and social science have independently arrived.

The hardware to wake up the neighbourhood up, not entirely unsurprisingly, the same as for a home but somewhat scaled up. A neighbourhood might decide to share a 20 foot battery container. Such a container can contain about 5 MWh of batteries and currently costs around 400 thousand euro's in Europe, including financing. Over 20 years and 200 households that works out to 10 euro's a month: less than one tenth of the reinforcement costs per household planned in the Netherlands. And it would give each household a whopping 25 kWh battery.

For a project developer constrained by a grid connection, the ability to build two or three times as many houses at a cost of 2000 euro per house would be a no-brainer. Furthermore the battery would already earn itself back by it's ability to buy the neighbourhood energy from outside when prices are low. If we extend the learning curve the same 20 foot container will cost less than 1000 euro per household in 2040 and my prediction is that every prudent and frugal neighbourhood will get one.

On top of this, grid operators should pay neighbourhoods for avoiding peaks on the grid because this substantially lowers grid reinforcement costs. A simulation we ran for the Dutch government at our company Zenmo quantified this. We modelled three control regimes for an ordinary Dutch street. First the scenario in which demand does not respond to supply at all. This leads to bad peaks, especially from electric vehicles. Secondly the scenario where electric vehicles, heat pumps, solar panels and batteries react to existing markets for electricity production. This lowered the highest peak of the year by 15%. Finally a scenario where avoiding the worst peaks was explicitly rewarded. It turned out surgically rewarding neighbourhoods for avoiding the worst peaks lowered the highest peak of the year by over 50%. And best of all: because these peaks only occur very seldom (e.g. 3% of the time) it doesn't take a lot of compensation to seduce end users.

We are not there yet. People working at the grid operators are still inventing legal and operational barriers. And that's the ones that are willing to talk about this at all. But once that hurdle has been taken, it makes sense that neighbourhoods become little federated kingdoms that manage most of their energy problems themselves. I think we have a chance because that way we would save a boatload of money and my fellow Dutchmen are very susceptible to that argument.

This is not just theoretical. Let me give some working examples, roughly in the order in which they came on line.

[Olympic Peninsula -- USA \(2007 to 2014\)](#)

One of the earliest field trials of neighbourhood-scale price coordination ran on a peninsula in Washington State. This was before the advent of solar panels, batteries, and electric vehicles. A hundred and twelve households used smart thermostats and water heaters and dryers, a five-minute double-auction market, and a strict constraint: a transformer that was not allowed to overload. Although flexibility was very limited compared to modern pilots, peak loads came down by 15 per cent while the participants reported no comfort loss. Source: [\(PNNL Olympic Peninsula final report\)](#).

[PowerMatching City -- Hoogkerk, the Netherlands \(2009 to 2015\)](#)

In a Groningen suburb called Hoogkerk, twenty-five Dutch households were wired up with PV, heat pumps, smart appliances and electric vehicles, all coordinated through a market that cleared every five minutes. The pilot grew to sixty households over five years. Peak load at the local transformer came down 30 to 35 per cent. The UN gave the project a sustainability award in 2012. The algorithm that ran it -- PowerMatcher -- had been published four years earlier by Koen Kok who is now a professor I often work with at my Eindhoven University of Technology. Sources: [\(Kok 2005\)](#); [\(PowerMatching City paper\)](#).

The solution has been staring us in the face for over 20 years but somehow the spell prevents us from recognizing it's value.

[SOLshare -- Bangladesh](#)

SOLshare is a fast-growing peer-to-peer trading platform for solar home systems. It connects 120 nanogrids serving more than 80 000 people, including a Rohingya refugee

camp. Households consume at different times of the day and -- due to solar panel orientation -- also generate at different times during the day. Trading between neighbours unlocks roughly thirty per cent of solar capacity that would otherwise be lost. Sources: ([2030.solutions profile](#)); ([Ashden Award](#)); ([UNFCCC Momentum for Change](#)).

Husk Power -- India and Nigeria

Husk is an initiative that's also growing fast. They claim to provide a "minigrid-in-a-box" solution that can be deployed within one day. In 2024 Husk doubled the number of minigrids it served from 200 to 400 and it now serves over 2 million people and 30 thousand businesses. Source: ([Husk Power news & insights](#)).

OMC Power -- India

OMC power has a unique and very clever holon solution that fixes the problem that rural cell towers for phone services often cannot get a grid connection. Without OMC, Telecom operators were running them on diesel generators. But refuelling was a pain and diesel was expensive, smelly and got stolen. Telecom operators hated paying for it since it could be up to [40% of their network operating cost](#). OMC's offer is simple: we'll build a solar plus battery plant and provide you all the electricity you need so you can stop buying diesel. Once OMC has an agreement with a Telecom operator, they can go to a bank to loan the money to build the plant and once they have that plant they can make it a little bit bigger and sell the excess electricity to houses near the cell tower. Is that clever or what? Source: ([OMC Power](#)).

It's actually very similar to how the electricity grid got started in Western counties roughly 120 years ago. Factories owners were often the launching customers and the electrification of households in the vicinity of the factory followed.

Sabang -- Palawan, the Philippines (operational since August 2019)

In the Philippines, there is a village called Sabang that sits at the entrance of one of the longest underground rivers in the world. It's a UNESCO heritage site that draws hundreds of thousands of visitors per year interested in the river or in diving. (Or in picking up young girls for money, but let's not spoil the mood.) And yet, until August 2019, each resort had to run its own diesel generator and households had no electricity at all. Now Sabang largely runs on renewable energy and households have electricity. What took so long was not the technology or the business case but cooperation of the local grid operator. That grid operator had an exclusive claim on Sabang but could not profitably connect it to the bigger grid. It took many years before it was willing to give up its claim. This is a story we see everywhere: incumbents unable to deliver but unwilling to give up claims are often the biggest hurdle for establishing local holons. Source: ([Microgrid Knowledge: CleanGrid Philippines / Sabang](#)).

Schoonschip -- Amsterdam (operational since 2021)

In a former harbour in Amsterdam, forty-seven houseboats have been quietly running their own grid for over 5 years now. About 500 solar panels divided across the roofs and thirty water-source heat pumps drawing low-grade warmth from the canal provide them with

most of the energy they need. With shared batteries and blockchain-mediated peer-to-peer energy trading they have reduced their grid connection tenfold. Their community started as a pilot that was condoned but will now become in violation of Dutch regulation. It's one of the clearest examples of how backward Dutch regulation is compared to European regulation with regard to holons. Tragically enough there is no defensible upside. Just sluggishness from the incumbents that are comfortably sleeping under the spell. Source: ([Metabolic project page](#)).

This is really happening

I could go on. About Rwanda using local energy cooperatives that are perfect examples of neighbourhood holons as the vehicle for rolling out home holons. Source: ([Rwanda Energy Group: off-grid electrification](#)). [whether the 48% target was hit on time] Or about Indonesia, that string of pearls, that archipelago of more than 17 thousand islands, where island after island is exchanging diesel generators for microgrids running on solar panels, batteries and grid-forming inverters. Source: ([IEEFA Indonesia 100 GW de-dieselisation](#)). The pattern is the same everywhere, no matter what country or continent. Holons running on renewable energy work and make neighbourhoods quieter and cleaner, while energy becomes reliable and affordable.

Holons all the way up

Do you remember Tempus and Hora? The two watchmakers? Tempus build his watch in one go, which was efficient, but easy to disrupt. Hora build his watch using subassemblies which was a bit more work but hardly suffered when disruptions happened. Herbert Simon created the parable to show why complex systems that survive consist of near-decomposable parts we call holons.

The important thing to understand about holons, is that it applies to a specific design pattern, not a specific size. We can create holons on the level of a house, or a neighbourhood, or a city, or a province, or a country, or a continent, or a planet. The essence of the design pattern we call an energy holon is that it's a unit that organises its internal energy flows optimally, with minimal data and energy exchange with the environment, while possibly containing smaller holons inside of it and being part of larger holons. The reason to create such a nested system of holons is not just to create a more resilient energy system, but also that it needs less electricity grid.

So your home energy management system (HEMS) is the energy brain of your house, choosing when to charge the electric car, use the heat pump, discharge the battery, curtail the solar panels, et cetera. It takes into account all the appliances, preferences etc. that are important in your home. When your neighbourhood holon gives your HEMS a few simple signals that indicates how the neighbourhood is doing, your HEMS can help to optimise the neighbourhood holon. For your neighbourhood holon this simplifies things because it doesn't need to know what's happening in your house in detail: your HEMS takes care of that. For you it's nice that you stay in control of your house and that it keeps working when the neighbourhood fails.

But we can repeat this pattern endlessly. For example: a group of neighbourhood holons could make up a district, a group of districts could make up a municipality, and a group of municipalities could make up a country. Each holon would handle 99% of energy decisions and most energy flows internally, while supporting the smaller holons inside it and the larger holon it was a part of.

On a country level, every country in the world could strive to become fully energy independent! Imagine that: no more problems or expenses getting fossil fuels from other countries! Trading some energy with other countries would still be mutually helpful, but it would be an option, not a necessity.

For me, primed by the first oil crisis and now irritated by our dependence on oil and gas from Russia, Saudi Arabia and the US, that seemed an extremely worthwhile goal. But is it achievable? Can the Netherlands really rid itself of its dependence on foreign oil and gas?

Making the Netherlands energy independent

The Netherlands uses a *lot* of energy. The table below shows how much.

Dutch energy use per sector in kWh per household per day	2023	Electrified future
Home electricity	5	6
Home heating	34	8
Inland mobility	50	15
Chemicals & pharmaceuticals	64	55
Oil refineries	14	3
Steel	10	2
Food industry	10	5
Building materials and other industry	9	8
Services and public buildings	29	15
Data centres	2	8
Greenhouses and other agricultural excl. mobility	13	5
Seasonal storage overhead	0	11
Subtotal without bunker fuels	240	139
International shipping	48	54
International aviation	16	37
Subtotal bunker fuels	64	91
Total	304	231

As you can see, in the current situation household electricity is relatively tiny while mobility and industry together make up three quarters of energy use. That's mostly because the Netherlands has a huge harbour (Rotterdam) that services a large part of Europe, a huge industry in chemicals, pharmaceuticals and refineries, and a large international airport. So most of our energy use is actually for international services. (Most

of our land use is agriculture which is also mainly for export by the way. The Dutch are very much into export.)

Let's assume that we electrify absolutely everything. The result of that is visible in the last column of the table. What jumps out is how much more efficient heating and inland mobility have become. I've assumed that melting ore to produce raw iron will be done in places where electricity is cheapest and few people live. You can also see that we invest 11 kWh per household every day in seasonal storage. This makes sure that we have enough energy to survive the long cold sun poor Dutch winters. All in all our energy use goes *down* from 240 to 139 kWh per household per day.

Now I've made one big crucial assumption here. I've assumed we will keep importing the fuel we need for international shipping and aviation. From a resilience standpoint that seems defensible to me. Firstly because something that interrupts fuel shipments will probably also interrupt other shipments and it's no use to produce fuel if the ships that need it are not arriving. Secondly, in a truly worst case situation, not getting fuel for international shipping and aviation would not cause distress inside the Netherlands and as such is more a question of international trade than national resilience. If you disagree you could allocate more land and/or sea to producing that fuel. It could very well be done but it's probably not economical.

So now we know what we need. But how much can we produce?

Solar and Wind in the Netherlands

The Netherlands is a fairly densely populated country with a little over 500 people per km², comparable to India, South Korea and Taiwan. The average household size is just above 2 people.

If we look at household holons, we end up with about 4000m² land per household. Furthermore these households can lay claim to around 1000m² of inland lakes and waterways and 7000m² of sea per household. Is that enough to become energy independent?

Let's cover **roof surfaces and parking spaces** with solar panels first. Research indicates that around 50% of roof surface, 30% of facades and 63% of parking could be used for solar panels. Together that's 171m² of solar panels per Dutch household. Notice this is a minute part of total land area of 4000m² and only surface that has no other useful function. Taking into account shading and positioning and long term 30% efficient panels, this could produce around 120 kWh per household per day.

Furthermore, we could place wind turbines, spread out over **10% of the North Sea**. The little artificial islands per wind turbine, spaced about 1.5 km apart, would only cover about 0.01% of the North Sea and actually *increase* biodiversity. The resulting wind would bring in 60 kWh per household per day, which is especially welcome in dark nights and Dutch winters that have relatively little sun. The total of these two measures lands at **180 kWh per household per day**.

That is actually *more* than what we need! We could electrify the Netherlands 100% with solar panels on unproductive surfaces and wind turbines spaced over 10% of the North Sea!

Where do we go from here?

We've now seen the holon stack from the bottom up. A house in rural Africa with four solar panels and a yellow lunch box that can evolve to a farmer completely escaping energy poverty. A street in Hoogkerk where appliances have been negotiating prices with each other every five minutes for over twenty years. A houseboat community in Amsterdam that has cut its grid connection by an order of magnitude. A village in Palawan where dive resorts and households share a microgrid as soon as the local incumbent stepped out of the way. A country -- the Netherlands -- that could meet its full electrified energy demand from rooftops and a few percent of the North Sea, with energy to spare.

All of this is based on proven technology. All of this will become increasingly logical as learning curves make renewable technology better.

So why are holons still the exception to the rule?

Because the rule is the problem. What's missing is permission. The regulatory ecosystem is still enforcing the grid design that those engineers in Berlin and Pittsburgh and Manchester came up with over a century ago. We are still under their spell. There are new solutions that could give us a cheaper and more resilient grid if we would only let them. Instead we are mowing them down before they can flower. That needs to change.